

# Image Captioning Task with Transformers [C5-Week4]

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## Abstract

*In the previous week, we realized classic recurrent pipelines severely underfit the complexity of human language and achieved bad results. Taking as a starting point the limitations observed, in this work, we evaluate more complex architectures, starting by an encoder-decoder architecture coupling ViT with GPT-2, and then scaling up to MLLMs, using Qwen for zero-shot evaluation and a custom pipeline performing PEFT using LoRA. We discover that jumping to transformers is the first big improvement, and specially MLLMs showed that massive pretraining is extremely powerful, although this potential can generate some problems if not taken care of correctly, provoking hallucination of the image content or deviating from what is really happening in the image. Ultimately, this work demonstrates what configurations and fine-tuning strategies are actually profitable for complex real-world datasets like VizWiz.*

## 1. Introduction

Since the invention of deep learning, it has been evolving in complexity and precision through the years, enabling powerful techniques that allow us to perform tasks that sit at the intersection of Computer Vision (CV) and Natural Language Processing (NLP). A specific case is Image Captioning, the process of automatically generating a natural language description from an image.

In the previous week, we started with the image captioning task using classic Deep Learning (DL) solutions. We explored baseline encoder-decoder architectures, specifically testing GRU and LSTM decoders combined with different word-level tokenizations. Although these structural changes provided some improvements, we realized these kinds of recurrent pipelines generally struggle to capture the full context of the images, showing a tendency to severely underfit the complexity of human language, and in general, obtain poor results.

Taking as a starting point the limitations observed, this week we aim to explore more advanced methodologies based on Transformers and Vision-Language Models

(VLMs) using the same dataset as last week, the VizWiz dataset, which is specifically designed to reflect real-world accessibility challenges with blurry and noisy images.

To achieve this, our work and contributions are divided into two main tasks. First, we evaluate the performance of an encoder-decoder architecture coupling ViT with GPT-2. We start with a zero-shot evaluation on VizWiz to have a reference baseline. Then, we try different strategies of fine-tuning (full fine-tuning, captioner fine-tuning, and backbone fine-tuning) to see how we can best adapt the model to our complex dataset.

Second, we perform image captioning, scaling up to Multimodal Large Language Models (MLLMs). Because of regional limitations with LLaMA, we opted for the Qwen family as our main captioner. We start by individually evaluating the zero-shot capabilities of different Qwen models, testing basic and advanced prompt engineering to extract deeper contextual understanding without updating weights. Finally, we build a custom pipeline performing Parameter-Efficient Fine-Tuning (PEFT) using LoRA. We connect our best ViT encoder from the first task into a smaller Qwen decoder, using a trainable MLP projection layer to translate the feature space, and train just the LoRA adapters to adapt the model efficiently.

Ultimately, the idea is that this investigation acts as a comprehensive exploration of the evolution from traditional CNN-RNN pipelines to modern multimodal architectures, testing what configurations and fine-tuning strategies are actually profitable for complex real-world datasets.

## 2. Related Work

Image captioning has evolved over the past decade. This section reviews the primary methodologies relevant to our work, showing the transition from recurrent models, which were the focus of the previous week, to modern multimodal architectures, as we will try in this project.

### 2.1. From Recurrent Models to Transformers

The modern era of image captioning started with classic DL solutions using an encoder-decoder architecture. The Show and Tell model [16] established this paradigm by uti-

lizing deep CNN features as input to an LSTM decoder. Later, Show, Attend and Tell [17] addressed the limitations of a single static feature by introducing attention mechanisms, allowing the decoder to focus on different spatial regions. However, later it was highlighted that these recurrent CNN+RNN pipelines generally struggle to capture the full global context of the images and have a tendency to severely underfit the complexity of human language [14][18]. Because of this, the field shifted towards Transformers and Vision Transformers (ViT) [15] [4] and ended up becoming the SOTA since self-attention mechanisms allow processing the image as a sequence of patches in parallel, significantly boosting generalization compared to recurrent pipelines.

## 2.2. Modular Vision-Language Architectures

After the success of Transformers, instead of training full networks from scratch, researchers started coupling strong pre-trained vision encoders with strong language decoders. Architectures like CPTR [9] demonstrated that full transformer networks (e.g., coupling a ViT with a GPT-2 decoder) worked exceptionally well. This modular approach is very important because it allows for easy plug-and-play with different models depending on the computational limits. In this case, the ViT extracts dense visual semantics, and these representations are directly fed into the input space of the language decoder.

## 2.3. Bridging Modalities in Large Language Models

Following this idea, the evolution arrived at Multimodal Large Language Models (MLLMs). As the recent survey by Abdulgalil and Basir [1] points out, we are now in a next-generation phase where massive models originally designed just for text are adapted to see. A clear example of this is the LLaVA architecture [8]. Instead of complex cross-attention mechanisms like in earlier models (e.g., BLIP [6, 7]), LLaVA simply uses a frozen ViT, a trainable projection layer (a simple MLP) to translate the feature space, and a large LLM. This way, the language model can read the visual embeddings as if they were text tokens, leveraging its immense pre-trained capacity for complex reasoning.

## 2.4. Parameter-Efficient Fine-Tuning (PEFT)

Although MLLMs achieve incredible results, fine-tuning all their parameters for a specific downstream task (like adapting them to the blurry and particular images of VizWiz) is completely prohibitive for standard computational resources. This is where Parameter-Efficient Fine-Tuning (PEFT) comes into focus. The most representative technique is LoRA (Low-Rank Adaptation) [5], which freezes the pretrained model weights and injects trainable low-rank matrices into specific layers. In modern LLM captioning, it is a common practice to apply these LoRA adapters specifically to the attention modules ( $Q_{proj}$ ,  $V_{proj}$ ,  $K_{proj}$ ,  $O_{proj}$ )

of the decoder. This ensures that we can adapt the vocabulary and style of the model to complex datasets while maintaining computational efficiency and preventing catastrophic forgetting of the base model.

# 3. Methodology

## 3.1. Previous Week

In the previous week, we started with the image captioning task using classic DL solutions. We adopted a progressive methodology to incrementally update our initial baseline. This baseline originally consisted of a ResNet-18 encoder and a GRU decoder operating with character-level tokenization. Through systematic experimentation, we explored the impact of scaling the encoder to a ResNet-34, comparing the GRU and LSTM decoder architectures, and modifying the tokenization level to evaluate subword and word representations. Furthermore, we analyzed the effect of introducing Bahdanau attention and conducted a final hyperparameter search. Our results demonstrated that the most significant performance improvements came from structural changes, specifically the transition to higher token levels like words, rather than just hyperparameter tuning. Ultimately, the experiments also highlighted a general tendency of these recurrent models to severely underfit the complexity of human language. Compared to other groups, we realized that we achieved bad results, for this reason we expect to significantly improve the results with transformers.

## 3.2. Task 1: ViT-GPT2 Architecture

First, we're going to evaluate the performance of an encoder-decoder architecture using the NLP Connect architecture from Hugging Face which couples ViT-B16 with GPT2 for captioning tasks.

Then, we're going to try different strategies of fine-tuning in order to adapt the model to our complex dataset, which, as previously commented, significantly differs from other datasets because of its particular nature. The fine-tuning strategies tested were a full fine-tuning on the model, fine-tuning with the backbone freeze (Captioner fine-tuning), and fine-tuning with the captioner freeze (backbone fine-tuning).

Due to computational limits, we did not perform a parameter search, although we tested different training configurations detailed in Section 4.

After doing these experiments, we also tested with other pairs of encoder-decoders for captioning with the best configuration found in the previous fine-tuning with ViT-GPT2. We tried BEiT-GPT2 [3, 10], ViT-DistilGPT2 [4, 13], ViT-SmolLM [2, 4], ViT-T5 [4, 12], and CLIP-GPT2 [10, 11].

### 3.3. Task 2: LoRA-tuned LLMs

Now we’re going to do image captioning with LLMs and MLLMs. For this, we’ll use Qwen since regional limitations don’t allow us to work with LLaMA. Specifically, we’re going to try three different Qwen architectures: Qwen2.5-VL-7B-Instruct, Qwen3-VL-8B-Instruct, and Qwen3.5-9B.

As before, the first step is to individually evaluate each model by itself, for this, prompt engineering is key in order to ask the model to produce adequate captions. For this, we tried two prompts, one basic prompt and another advanced, which was more complete and included a user prompt as well as a system prompt.

Then, we performed PEFT using LoRA to Qwen3.5-0.8B and Qwen3.5-4B, which we considered the equivalents of LLaMA requested on the original instructions. The pipeline here consists of the best backbone of Task 1, a trainable projection layer, and a LoRA Qwen model. This projection layer is a trainable MLP and is necessary because the encoding of the backbone differs from the Qwen one, and therefore, a translation of the feature space is necessary. Finally we perform LoRA to the attention modules  $Q_{proj}$ ,  $V_{proj}$ ,  $K_{proj}$ ,  $O_{proj}$ .

Then, we perform a sequential hyperparameter search on both models, first we search for  $r$  and  $\alpha$ , then we try different learning rates because we hypothesized that the PEFT will allow for faster training as the number of parameters to train isn’t very high, and finally, we will test different dropout rates in order to refine the training.

### 3.4. Evaluation Metrics

The quantitative metrics used for evaluation are the following:

- **BLEU-1 and BLEU-2:** These are two variations of the same metric, with the first one checking if every predicted word appears on the original captions, and the second one checking for pairs of words ( Unigrams vs Bigrams). BLEU-1 can fail when the model uses the correct words in a meaningless order, so it’s important to monitor both of them.
- **ROUGE-L:** This metric looks at how similar the phrases are by checking what the longest shared sequences of Unigrams are. They do not have to be consecutive, but they do have to be in the same order. This metric is more strict on how word order should occur, such as an adjective followed by a noun, whilst not letting the model add things like another adjective to the noun without being highly punished.
- **METEOR:** This metric allows even more flexibility, incorporating two steps, stemming and synonym matching, that allow matches between words that are written slightly differently or have very similar meanings.

Qualitatively, we fixed multiple examples in the test partition and analysed these examples. Here, the most interest-



Figure 1. Example of image from VizWiz Dataset

ing examples are reported, but the qualitative evaluation is consistent across all the experiments.

## 4. Experimental Settings

### 4.1. Dataset

The dataset used is the VizWiz-Captions Dataset. This dataset contains a series of images taken by blind people and their corresponding captions. Because these images are taken by blind people, a lot of images are out of focus or blurry, making the challenge more difficult.

The dataset includes 23431 training images, 7750 validation images and 8000 test images. For each image, there are 5 different captions. Because we do not have access to the test captions, we instead used the validation images as our test set.

### 4.2. Implementation Details

We did all our inference and training on two different clusters. We used the cluster provided by the university with 3090 GPUs, as well as a personal cluster with Ada 6000 and L40s GPUs.

For the first task, we used the HuggingFace Transformers library along with PyTorch, for the second task we used these same libraries but also included the PEFT library for training our Lora. We logged everything using WandB. In order to decide the best model we select the model with the best metrics at the epoch with less validation loss.

### 4.3. Settings for Task 1

For fine-tuning we didn’t do a big hyperparameter search due to computational limitations, but we decided the hyperparameters based on prior knowledge working on similar problems. The main considerations on the configurations tested was that as bigger was the model in general the LR had to decrease, arriving to a  $1 \cdot 10^{-5}$  for the full fine tuning,  $2 \cdot 10^{-5}$  for the ViT and  $5 \cdot 10^{-5}$  for GPT-2. For all these runs we used AdamW as the optimizer. We also use a linear warmup of 1000 steps and a batch size of 16.

We also experimented with a slight data augmentation during training, specifically applying mild random affine

transformations (just 5 degrees of rotation and a 5% translation) and horizontal flips. We did this because the dataset itself is already difficult and adding more heavy distortions to the image would make the task too hard; we basically tried to simulate the natural hand jitter of the users taking the photos.

For the generation part, we used beam search with a number of 4 beams and set a constraint of no repeating ngrams of 3 or more. We trained the models until convergence.

#### 4.4. Settings for Task 2

For the basic prompt we didn't add a system prompt and the user prompt was "Describe this image briefly.". For the advanced setting, we added a detailed system prompt to give the model more context about the problem: "You are an image captioning assistant helping visually impaired people understand their surroundings. Images may be blurry, poorly lit, or taken at unusual angles. Describe what you see concisely and factually, focusing on the main subject.". Together with this, the user prompt was way more restrictive: "Describe this image in one short sentence. Output only the caption, nothing else.". For the generation in this zero-shot evaluation we use greedy decoding.

For the PEFT fine-tuning with the custom pipeline, we used the AdamW optimizer and a batch size of 8. Since we were only training the LoRA adapters and the projection layer, we reduced the linear warmup to just 100 steps and trained until 5 epochs because of the high execution time.

As mentioned in Section 3 we did and hyperparameter search, for the rank  $r$  we tried 8 and 16, and for  $\alpha$  16 and 32. For the LR we tested  $5 \cdot 10^{-4}$ ,  $1 \cdot 10^{-4}$  and  $5 \cdot 10^{-5}$  and the values for the dropout tested were 0.05, 0.025 and 0.01. As mentioned in the methodology, we targeted the specific attention modules of the Qwen decoder, applying the adapters to the  $Q_{proj}$ ,  $V_{proj}$ ,  $K_{proj}$ ,  $O_{proj}$ . Finally, we kept the exact same slight data augmentation strategy that we used in Task 1 for the training images.

## 5. Results

### 5.1. Task 1: Encoder + Decoder Architecture

#### 5.1.1. Zero Shot Evaluation

Initial results without fine-tuning showed the model had some descriptive ability<sup>1</sup>, indicated by the high BLEU-1 score, but wasn't correctly describing the scene as seen in the meteor score.

Qualitative results show that the model got general objects and colors right, but was struggling with the semantics of the object. It recognized bottles, but not the content inside the bottle.

Another big problem came with "caption-less images". These are images that are either blurry or contain no ob-

jects, so the given captions just say "Quality issues are too severe to recognize visual content", but the model still tries to predict something.

Table 1. Zero Shot Evaluation of the ViT+GPT2

|          | METEOR | ROUGE-L | BLEU-1 | BLEU-2 |
|----------|--------|---------|--------|--------|
| Baseline | 0.283  | 0.367   | 0.542  | 0.340  |

#### 5.1.2. Multiple Fine-tuning Strategies

Fine-tuning simply the captioner showed the best results<sup>2</sup>, only losing to the baseline on the BLEU-1 score. Training the backbone seems to cause a big loss in both BLEU scores, and we are not entirely sure where it is coming from.

It was nice to see that, as we predicted, training allowed the model to caption images as too blurry to caption. Though it is interesting to see that only the caption fine-tuning acquired this ability.

Qualitatively, we saw that for some reason, the models started getting confused with colors, labeling a blue cloth as both purple and brown.

Table 2. Results of various fine tuning strategies

| Model     | METEOR       | ROUGE-L      | BLEU-1       | BLEU-2       |
|-----------|--------------|--------------|--------------|--------------|
| Baseline  | 0.283        | 0.367        | <b>0.542</b> | 0.340        |
| Captioner | <b>0.424</b> | <b>0.429</b> | 0.495        | <b>0.362</b> |
| Backbone  | 0.420        | 0.356        | 0.278        | 0.208        |
| All       | 0.414        | 0.357        | 0.270        | 0.202        |

We would also like to mention that we tried other architectures, BEiT-GPT2, ViT-DistilGPT2, ViT-SmolLM, ViT-T5, and CLIP-GPT2, using the best hyperparameters for ViT-GPT2, but none of them outperformed our current baseline.

#### 5.1.3. Other Transformer Models

After testing and fine-tuning the ViT+GPT2 model, we wanted to try some different captioning models. We tried pairing different backbones with GPT2, changing GPT2 for other captioning modules, as well as the GIT model. The results can be observed in Table 3.

Table 3. Comparison of different Transformer models and backbones.

| Model          | Meteor       | Rouge-l      | Bleu-1       | Bleu-2       |
|----------------|--------------|--------------|--------------|--------------|
| ViT+GPT2       | <b>0.424</b> | <b>0.429</b> | <b>0.495</b> | <b>0.362</b> |
| BEiT+GPT2      | 0.27         | 0.26         | 0.26         | 0.18         |
| ViT+DistilGPT2 | 0.125        | 0.129        | 0.05         | 0.143        |
| GIT            | 0.16         | 0.32         | 0.36         | 0.118        |

Unfortunately, none of these tests gave comparable results to the ViT+GPT2 model. Changing the backbone



or captioner makes the fine-tuning process harder, as the two halves have to adapt to each other whilst learning the new dataset. As for GIT, we believe we may have made mistakes on implementation, as the results for a pretrained model shouldn't have been that low, but we didn't want to spend more time when our previous fine-tune had already achieved good results. With more testing and time, we believe they could have achieved similar, if not better, scores.

## 5.2. Task 2: MLLM Captioning

### 5.2.1. Zero Shot Captioning

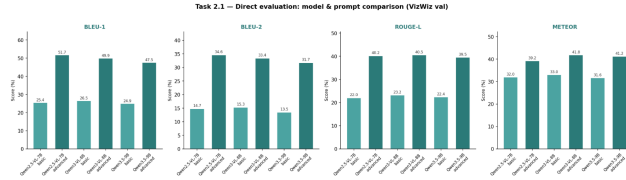


Figure 2. Captioning Metrics for different versions of Qwen on different prompts

As previously commented, we evaluated Qwen2.5-VL-7B, Qwen3-VL-8B, and Qwen3.5-9B for the Zero Shot test. As seen in Figure 2, the three models achieve similar performance across all metrics, so we cannot identify a clear winner considering all metrics. However, we can see that all models benefit from using an advanced prompt instead of a basic one, with differences up to 50% in some cases. In general, we observed that this is due to the advanced prompt forcing shorter sentences, like the ones found on the dataset, with the basic usually generates longer sequences.

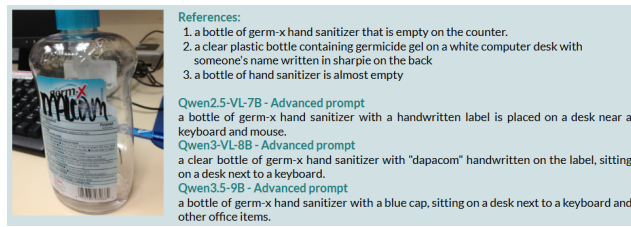


Figure 3. Qualitative examples for Zero Shot on Qwen

Focusing on METEOR (what we consider the most robust metric as it accounts for synonyms and paraphrasing), Qwen3-VL-8B achieves the best result (41.8), followed closely by Qwen3.5-9B (41.2) and Qwen2.5-VL-7B (39.2). This predominance of Qwen3 is also maintained in ROUGE-L, but when we take a look at the BLEU metrics, the smaller the model, the better. After taking a look at qualitative results, we think this happens because the bigger models are more expressive and rarely give straightforward

and simple descriptions, using complex words or sophisticated formulations for the description. This can be observed in Figure 3 where the bigger models focus on more details.

### 5.2.2. LoRA PEFT

For the experimentation using LoRA in Qwen3.5-0.8B and Qwen3.5-4B, we performed a hyperparameter search of  $r$  and  $\alpha$ , then for the LR, and finally for the dropout values.

| Model        | r  | $\alpha$ | Meteor       | Rouge-l      | Bleu-1       | Bleu-2       |
|--------------|----|----------|--------------|--------------|--------------|--------------|
| Qwen3.5-0.8B | 8  | 16       | <b>0.311</b> | 0.196        | 0.200        | <b>0.131</b> |
|              | 8  | 32       | 0.298        | <b>0.209</b> | 0.204        | 0.129        |
|              | 16 | 16       | 0.255        | 0.179        | <b>0.208</b> | 0.120        |
|              | 16 | 32       | 0.289        | 0.189        | 0.195        | 0.124        |
| Qwen3.5-4B   | 8  | 16       | <b>0.355</b> | <b>0.278</b> | <b>0.249</b> | <b>0.170</b> |
|              | 8  | 32       | 0.338        | 0.259        | 0.241        | 0.157        |
|              | 16 | 16       | 0.304        | 0.247        | 0.218        | 0.147        |
|              | 16 | 32       | 0.334        | 0.272        | 0.239        | 0.158        |

Table 4. LoRA Hyperparameters Evaluation.

After performing the PEFT with different LoRA configurations, we obtained the results shown in Table 4. The best configuration for both models was  $r=8$ ,  $\alpha=16$ . We think lower rank generalizes better on the relatively small VizWiz training set, suggesting that higher capacity leads to overfitting rather than improvement.

Comparing the best LoRA configuration ( $r=8$ ,  $\alpha=16$ ) across both decoder sizes, Qwen3.5-4B outperforms Qwen3.5-0.8B on all metrics, with a METEOR improvement of +4.4 points (35.5 vs 31.1).

Given this consistent advantage, we select Qwen3.5-4B as our decoder for all the subsequent experiments on learning rate and dropout.

After that, we also conducted experiments on the LR and the Dropout probabilities, the results are shown in Figure 4. Since the epochs are limited because of the computational resources, the runs with a higher learning rate show better results for most of the metrics, but at the highest LR, it starts to struggle as can be seen on the METEOR metric. For the dropout, it seems that a smaller value works best, probably because the work was already distributed between neurons in the original training.

With this final model and training configurations, we still obtain worse results than when zero-shotting, but this is because we're not applying any post-processing to the model. Applying a repetition penalty and removing chat template artifacts produces a much cleaner caption, closer to the references, as can be observed in Table 5. The output is concise and demonstrates that the model has learned a useful visual-text mapping during fine-tuning.

However, it remains less precise than the zero-shot results from Task 2.1, where the model was able to capture

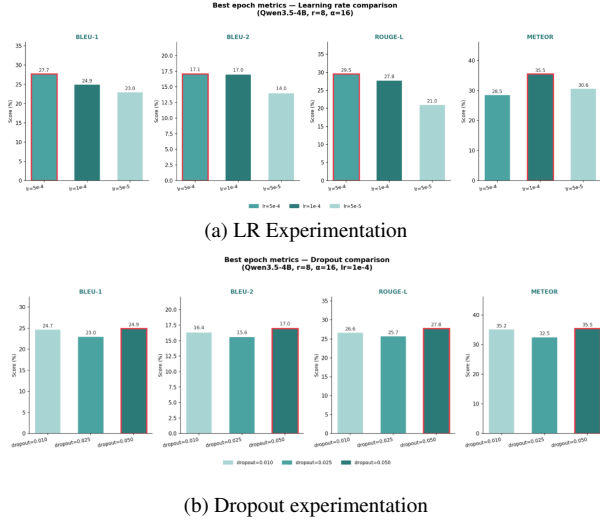


Figure 4. Results after trying different LR and Dropout values for Qwen3.5-4B

|           | METEOR | ROUGE-L | BLEU-1 | BLEU-2 |
|-----------|--------|---------|--------|--------|
| Best LoRA | 0.3817 | 0.4408  | 0.5728 | 0.4232 |

Table 5. Metrics with refined output for best LoRA

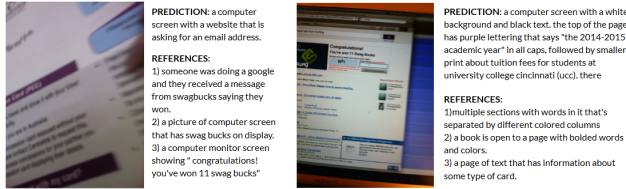


Figure 5. Qualitative Examples after LoRA

the details of the images. This suggests that while LoRA fine-tuning teaches the model the caption format, it does not completely compensate for the richer visual understanding of a trained multimodal model.

Finally, in Figure 5 we can observe some qualitative examples. It is interesting how sometimes the model hallucinates, as it happens in both examples, by thinking the webpage is asking for an email when it's not, or reading "2014-2015 academic year". Also is noticeable that for many cases we obtain really long captions.

### 5.3. Discussion and Comparisons

If we look at the general results summarized in Table 6, we can see a very clear evolution in performance. The first thing that stands out is the absolute failure of the classic CNN-RNN solutions from the previous week, lacking the capacity to capture any real language structure, mostly just

Table 6. Metric comparison by task and architecture

| Task          | Architecture | Meteor       | Rouge-l      | Bleu-1       | Bleu-2       |
|---------------|--------------|--------------|--------------|--------------|--------------|
| Previous Week | Best Model   | 0.196        | 0.286        | 0.201        | 0.136        |
| Task 1        | Baseline     | 0.283        | 0.367        | 0.542        | 0.340        |
|               | Captioner    | <b>0.424</b> | 0.429        | 0.495        | 0.362        |
|               | Backbone     | 0.420        | 0.356        | 0.278        | 0.208        |
|               | All          | 0.414        | 0.357        | 0.270        | 0.202        |
| Task 2        | Zero Shot    | 0.415        | 0.405        | 0.499        | 0.334        |
|               | LoRA         | 0.382        | <b>0.441</b> | <b>0.572</b> | <b>0.423</b> |

repeating common tokens without sense.

The jump to Transformers in Task 1 was the first big improvement. Just by using a ViT-GPT2 with full fine-tuning, we arrived at a BLEU-2 of 0.202 and a METEOR of 0.414, which is way better than anything we did before. This happened because the self-attention in the ViT handles the noisy images much better than a simple CNN, and GPT-2 has a much more robust understanding of how to build a sentence. In Task 2, when we first tested the MLLMs like Qwen3-4B in zero-shot, we saw that massive pretraining is extremely powerful, achieving a METEOR of 0.415 just with a good advanced prompt.

But the most interesting part is our LoRA fine-tuning. Although the "raw" LoRA output struggled with chat template artifacts, once we applied the refined post-processing and repetition penalty, the results improved in all metrics, losing only in METEOR to the ViT+GPT2, where the Captioner unfreezed architecture remains the best, probably caused by Qwen hallucinating, which sometimes causes the model to ignore what is really happening in the image, specifically after the LoRA fine-tuning which its architecture added the MLP that translates the features from the ViT to Qwen, making them imperfect and promoting this types of visual misalignments.

## 6. Conclusions

In conclusion, this project allowed us to perform a comprehensive exploration of the evolution from traditional CNN-RNN pipelines to modern multimodal architectures. As previously commented, the classic solutions severely underfit the complexity of human language, resulting in an absolute failure where models mostly just repeated common tokens without sense. The jump to Transformers coupling a ViT with GPT-2 was the first big improvement, as self-attention handles the blurry and noisy images of the VizWiz dataset much better. For this architecture, fine-tuning simply the captioner showed the best results, as it uniquely allowed the model to learn the specific dataset vocabulary and caption images as too blurry to caption.

Scaling up to MLLMs, we saw that massive pretraining is extremely powerful. The Qwen models achieved very strong zero-shot results just by using an advanced

prompt that forced the model to be brief and factual. Finally, our custom pipeline performing PEFT using LoRA on Qwen3.5-4B showed that we can efficiently adapt the vocabulary of the model. Although applying a refined post-processing to remove chat template artifacts produced a much cleaner caption and improved the metrics significantly, we realized the model sometimes hallucinates and deviates of what is really happening in the image. We hypothesize this is because the MLP projection layer makes the feature translation between the ViT and Qwen imperfect, promoting visual misalignments. Ultimately, this work demonstrates what configurations and fine-tuning strategies are actually profitable for complex real-world datasets.

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